
Chapter 10

Testing the Difference between Means, Variances, and Proportions

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- **10-1 Introduction**
- **10-2 Testing the Difference
between Two Means: Large
Samples**
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- **10-4 Testing the Difference between Two Means: Small Independent Samples**
- **10-5 Testing the Difference between Two Means: Small Dependent Samples**

Outline

- **10-6 Testing the Difference between Proportions**

Objectives

- **Test the difference between two large sample means using the z test.**
- **Test the difference between two variances or standard deviations.**
- **Test the difference between two means for small independent samples.**

Objectives

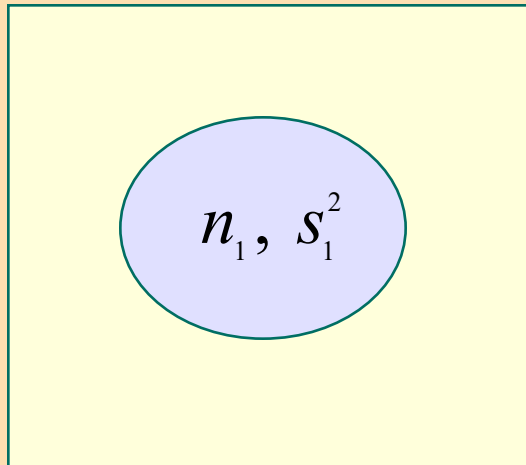
- **Test the difference between two means for small dependent samples.**
- **Test the difference between two proportions.**

10-2 Testing the Difference between Two Means: **Large Samples**

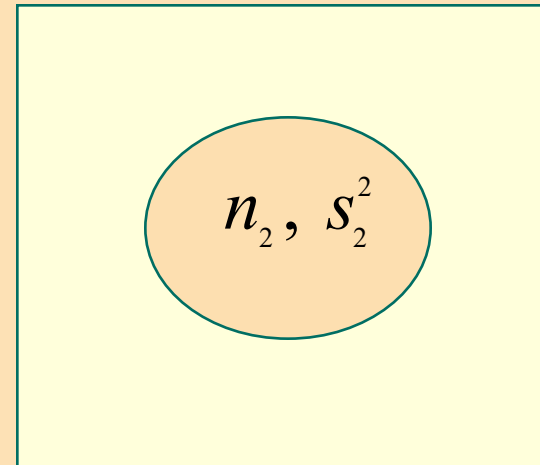
- **Assumptions for this test:**
- **Samples are independent.**
- **The sampling populations must be normally distributed.**
- **Standard deviations are known or samples must be at least 30.**

10-2 Testing the Difference between Two Means: Large Samples

$$\mu_1, \sigma_1^2$$



$$\mu_2, \sigma_2^2$$



10-2 Formula for the z Test for Comparing Two Means from Independent Populations

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

10-2 z Test for Comparing Two Means from Independent Populations - **Example**

- A survey found that the average hotel room rate in New Orleans is \$88.42 and the average room rate in Phoenix is \$80.61. Assume that the data were obtained from two samples of 50 hotels each and that the standard deviations were \$5.62 and \$4.83 respectively. At $\alpha = 0.05$, can it be concluded that there is no significant difference in the rates?

10-2 z Test for Comparing Two Means from Independent Populations - **Example**

- **Step 1:** State the hypotheses and identify the claim.
- $H_0: \mu_1 = \mu_2$ (claim) $H_1: \mu_1 \neq \mu_2$
- **Step 2:** Find the critical values. Since $\alpha = 0.05$ and the test is a two-tailed test, the critical values are $z = \pm 1.96$.
- **Step 3:** Compute the test value.

10-2 z Test for Comparing Two Means from Independent Populations - **Example**

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$
$$= \frac{(88.42 - 80.61) - 0}{\sqrt{\frac{5.62^2}{50} + \frac{4.83^2}{50}}} = 7.45$$

10-2 z Test for Comparing Two Means from Independent Populations - **Example**

- **Step 4:** Make the decision. Reject the null hypothesis at $\alpha = 0.05$, since $7.45 > 1.96$.
- **Step 5:** Summarize the results. There is enough evidence to reject the claim that the means are equal. Hence, there is a significant difference in the rates.

10-2 *P*-Values

- The *P*-values for the tests can be determined using the same procedure as shown in Section 9-3.
- The *P*-value for the previous example will be: $P\text{-value} = 2 \times P(z > 7.45) \approx 2(0) = 0$.
- You will reject the null hypothesis since the $P\text{-value} = 0 < \alpha = 0.05$.

10-2 Formula for Confidence Interval for Difference Between Two Means : Large Samples

$$\begin{aligned} & \left(\bar{X}_1 - \bar{X}_2 \right) - \left(z_{\alpha/2} \right) \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \\ & < \mu_1 - \mu_2 < \\ & \left(\bar{X}_1 - \bar{X}_2 \right) + \left(z_{\alpha/2} \right) \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \end{aligned}$$

10-2 Confidence Interval for Difference of Two Means: Large Samples - **Example**

- Find the 95% confidence interval for the difference between the means for the data in the previous example.
- Substituting in the formula one gets **(verify)** $5.76 < \mu_1 - \mu_2 < 9.86$.
- Since the confidence interval does not contain zero, one would reject the null hypothesis in the previous example.

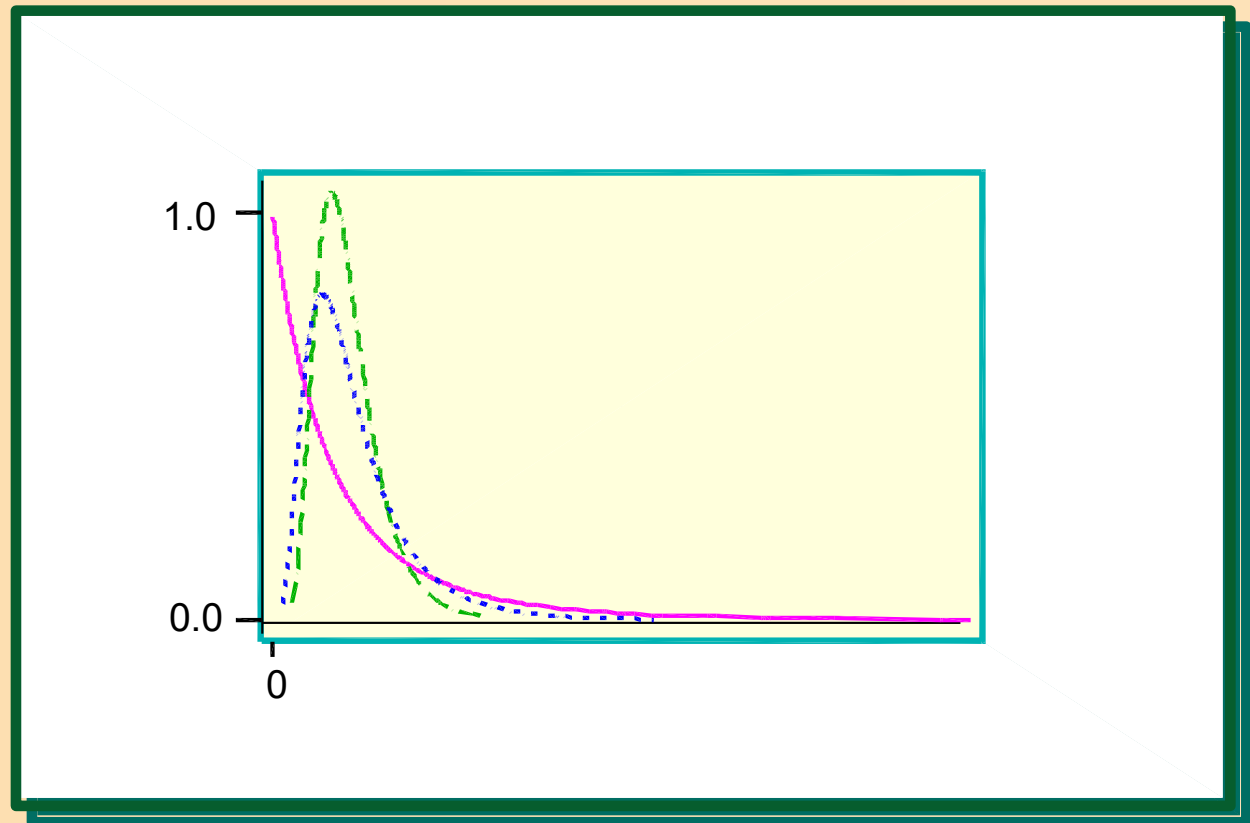
10-3 Testing the Difference Between Two Variances

- For the comparison of two variances or standard deviations, an ***F* test** is used.
- The sampling distribution of the variances is called the ***F* distribution**.

10-3 Characteristics of the F Distribution

- The values of F cannot be negative.
- The distribution is positively skewed.
- The mean value of F is approximately equal to 1.
- The F distribution is a family of curves based on the degrees of freedom of the variance of the numerator and denominator.

10-3 Curves for the F Distribution



10-3 Formula for the F Test

$$F = \frac{s_1^2}{s_2^2}$$

where s_1^2 is the larger of the two variances.

numerator degrees of freedom = $n_1 - 1$

denominator degrees of freedom = $n_2 - 1$

n_1 is the sample size from which the larger variance was obtained.

10-3 Assumptions for Testing the Difference between Two Variances

- The populations from which the samples were obtained must be normally distributed.
- The samples must be independent of each other.

10-3 Testing the Difference between Two Variances - **Example**

- A researcher wishes to see whether the variances of the heart rates (in beats per minute) of smokers are different from the variances of heart rates of people who do not smoke. Two samples are selected, and the data are given on the next slide. Using $\alpha = 0.05$, is there enough evidence to support the claim?

10-3 Testing the Difference between Two Variances - Example

- For smokers $n_1 = 26$ and $s_1^2 = 36$; for nonsmokers $n_2 = 18$ and $s_2^2 = 10$.
- **Step 1:** State the hypotheses and identify the claim.
 $H_0: \sigma_1^2 = \sigma_2^2$ $H_1: \sigma_1^2 \neq \sigma_2^2$ (claim)

10-3 Testing the Difference between Two Variances - Example

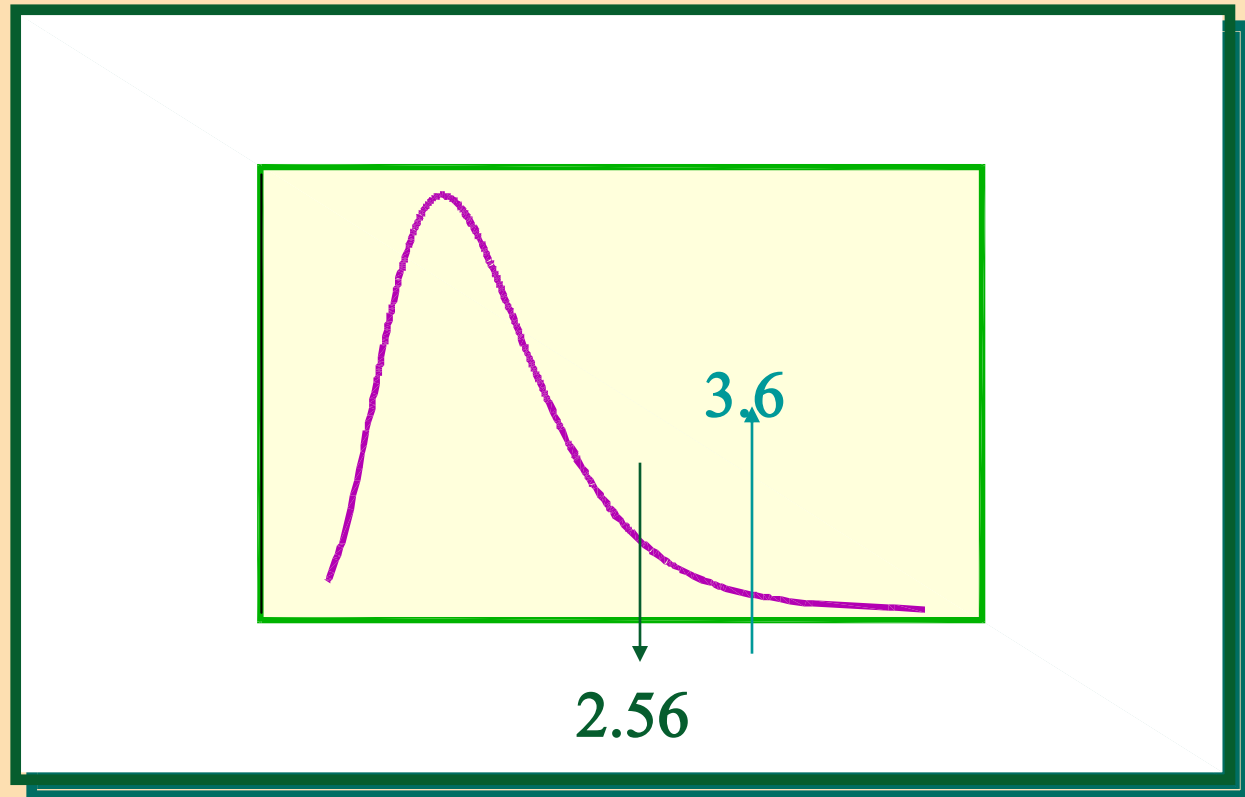
- **Step 2:** Find the critical value. Since $\alpha = 0.05$ and the test is a two-tailed test, use the 0.025 table. Here d.f. N. = $26 - 1 = 25$, and d.f.D. = $18 - 1 = 17$. The critical value is $F = 2.56$.
- **Step 3:** Compute the test value.
 $F = s_1^2 / s_2^2 = 36/10 = 3.6$.

10-3 Testing the Difference between Two Variances - **Example**

- **Step 4:** Make the decision. Reject the null hypothesis, since $3.6 > 2.56$.
- **Step 5:** Summarize the results. There is enough evidence to support the claim that the variances are different.

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10-3 Testing the Difference between Two Variances - Example



10-3 Testing the Difference between Two Variances - Example

- An instructor hypothesizes that the standard deviation of the final exam grades in her statistics class is larger for the male students than it is for the female students. The data from the final exam for the last semester are: males $n_1 = 16$ and $s_1 = 4.2$; females $n_2 = 18$ and $s_2 = 2.3$.

10-3 Testing the Difference between Two Variances - Example

- Is there enough evidence to support her claim, using $\alpha = 0.01$?
- **Step 1:** State the hypotheses and identify the claim.
 $H_0: \sigma_1^2 \leq \sigma_2^2$ $H_1: \sigma_1^2 > \sigma_2^2$ (claim)

10-3 Testing the Difference between Two Variances - Example

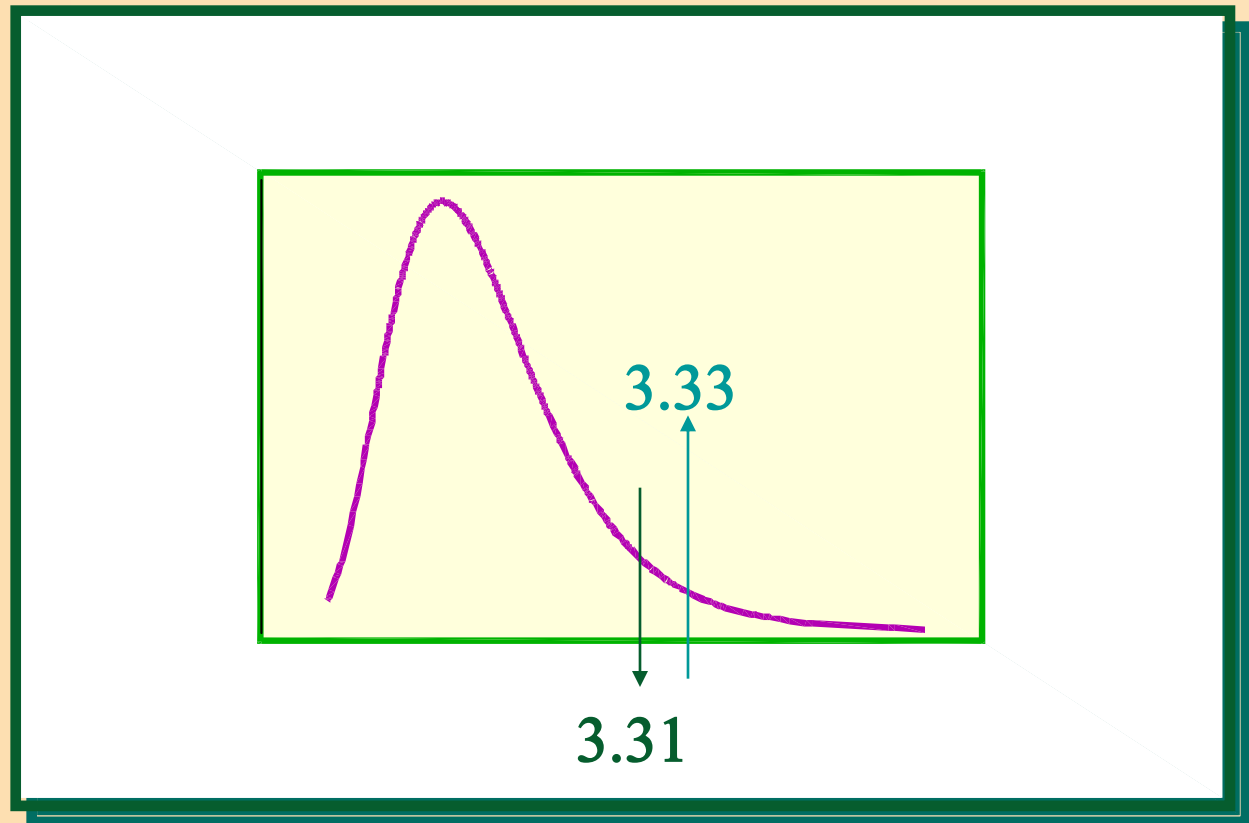
- **Step 2:** Find the critical value. Here, $d.f.N. = 16 - 1 = 15$, and $d.f.D. = 18 - 1 = 17$.
For $\alpha = 0.01$ table, the critical value is $F = 3.31$.
- **Step 3:** Compute the test value.
 $F = (4.2)^2 / (2.3)^2 = 3.33$.

10-3 Testing the Difference between Two Variances - **Example**

- **Step 4:** Make the decision. Reject the null hypothesis, since $3.33 > 3.31$.
- **Step 5:** Summarize the results. There is enough evidence to support the claim that the standard deviation of the final exam grades for the male students is larger than that for the female students.

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10-3 Testing the Difference between Two Variances - Example



10-4 Testing the Difference between Two Means: **Small Independent Samples**

- When the sample sizes are small (< 30) and the population variances are unknown, a t test is used to test the difference between means.
- The two samples are assumed to be independent and the sampling populations are normally or approximately normally distributed.

10-4 Testing the Difference between Two Means: **Small Independent Samples**

- There are two options for the use of the t test.
- When the variances of the populations are equal and when they are not equal.
- The F test can be used to establish whether the variances are equal or not.

10-4 Testing the Difference between Two Means: Small Independent Samples - Test Value Formula

Unequal Variances

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

d. f. = smaller of $n_1 - 1$ or $n_2 - 1$

10-4 Testing the Difference between Two Means: Small Independent Samples - Test Value Formula

Equal Variances

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$d. f. = n_1 + n_2 - 2.$$

10-4 Difference between Two Means: Small Independent Samples - **Example**

- The average size of a farm in Greene County, PA, is 199 acres, and the average size of a farm in Indiana County, PA, is 191 acres. Assume the data were obtained from two samples with standard deviations of 12 acres and 38 acres, respectively, and the sample sizes are 10 farms from Greene County and 8 farms in Indiana County. Can it be concluded at $\alpha = 0.05$ that the average size of the farms in the two counties is different?

10-4 Difference between Two Means: Small Independent Samples - **Example**

- Assume the populations are normally distributed.
- First we need to use the F test to determine whether or not the variances are equal.
- The critical value for the F test for $\alpha = 0.05$ is 4.20.
- The test value = $38^2/12^2 = 10.03$.

10-4 Difference between Two Means: Small Independent Samples - **Example**

- Since $10.03 > 4.20$, the decision is to reject the null hypothesis and conclude the variances are not equal.
- **Step 1:** State the hypotheses and identify the claim for the means.
- $H_0: \mu_1 = \mu_2$ $H_1: \mu \neq \mu_2$ (claim)

10-4 Difference between Two Means: Small Independent Samples - **Example**

- **Step 2:** Find the critical values. Since $\alpha = 0.05$ and the test is a two-tailed test, the critical values are $t = -2.365$ and $+2.365$ with d.f. = $8 - 1 = 7$.
- **Step 3:** Compute the test value. Substituting in the formula for the test value when the variances are not equal gives $t = 0.57$.

10-4 Difference between Two Means: Small Independent Samples - **Example**

- **Step 4:** Make the decision. Do not reject the null hypothesis, since $0.57 < 2.365$.
- **Step 5:** Summarize the results. There is not enough evidence to support the claim that the average size of the farms is different.
- **Note:** If the the variances were equal - use the other test value formula.

10-4 Confidence Intervals for the Difference of Two Means: **Small Independent Samples**

Unequal Variances

$$(\bar{X}_1 - \bar{X}_2) - t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$< \mu_1 - \mu_2 <$$

$$(\bar{X}_1 - \bar{X}_2) + t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

d. f. = *smaller of* $n_1 - 1$ *or* $n_2 - 1$

10-4 Confidence Intervals for the Difference of Two Means: **Small Independent Samples**

Equal Variances

$$\begin{aligned} (\bar{X}_1 - \bar{X}_2) - t_{\alpha/2} \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \\ < \mu_1 - \mu_2 < \\ (\bar{X}_1 - \bar{X}_2) + t_{\alpha/2} \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \end{aligned}$$

d. f. = $n_1 + n_2 - 2$.

10-5 Testing the Difference between Two Means: **Small Dependent Samples**

- When the values are dependent, employ a t test on the differences.
- Denote the differences with the symbol D , the mean of the population of differences with μ_D , and the sample standard deviation of the differences with s_D .

10-5 Testing the Difference between Two Means: **Small Dependent Samples** - Formula for the test value.

$$t = \frac{\bar{D} - \mu_D}{s_D / \sqrt{n}}$$

where

$\bar{D}$ = *sample mean*

degrees of freedom = $n - 1$

10-5 Testing the Difference between Two Means: Small Dependent Samples - Formula for the test value.

- **Note:** This test is similar to a one sample t test, except it is done on the differences when the samples are dependent.

10-5 Confidence Interval for the Difference between Two Means: **Small Dependent Samples - Formula.**

$$\bar{D} - t_{\alpha/2} \cdot s_D / \sqrt{n} < \mu_D < \bar{D} + t_{\alpha/2} \cdot s_D / \sqrt{n}$$

$$\text{d. f.} = n - 1$$

Note: This formula is similar to the confidence interval formula for a single population mean when the population variance is unknown.

10-6 Testing the Difference between Proportions - Formula

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\bar{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where n_1 and n_2 are sample sizes

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2}; \quad \bar{q} = 1 - \bar{p}; \quad \hat{p}_1 = \frac{X_1}{n_1}; \quad \hat{p}_2 = \frac{X_2}{n_2};$$

X_1 = number of successes in sample 1

X_2 = number of successes in sample 2

10-6 Testing the Difference between Proportions - **Example**

- A sample of 50 randomly selected men with high triglyceride levels consumed 2 tablespoons of oat bran daily for six weeks. After six weeks, 60% of the men had lowered their triglyceride level. A sample of 80 men consumed 2 tablespoons of wheat bran for six weeks. **(continued on next slide)**

10-6 Testing the Difference between Proportions - **Example**

- After six weeks, 25% had lower triclyceride levels. Is there significant differences in the two proportions, at the 0.01 level of significance?

10-6 Testing the Difference between Proportions - Example

$$\hat{p}_1 = 60\% = 0.60; \quad \hat{p}_2 = 25\% = 0.25;$$

$$X_1 = (0.60)(50) = 30;$$

$$X_2 = (0.25)(80) = 20;$$

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{30 + 20}{50 + 80} = 0.385;$$

$$\bar{q} = 1 - 0.385 = 0.615.$$

10-6 Testing the Difference between Proportions - Example

- **Step 1:** State the hypotheses and identify the claim.
- $H_0: p_1 = p_2$ $H_1: p_1 \neq p_2$ (claim)
- **Step 2:** Find the critical values. Since $\alpha = 0.01$, the critical values are +2.58 and -2.58.
- **Step 3:** Compute the test value.
 $z = 3.99$ (verify using the formula).

10-6 Testing the Difference between Proportions - Example

- **Step 4:** Make the decision. Reject the null hypothesis, since $3.99 > 2.58$.
- **Step 5:** Summarize the results. There is enough evidence to support the claim that there is a difference in proportions.

10-6 Confidence Interval for the Difference between Two Proportions

$$\begin{aligned} & (\hat{p}_1 - \hat{p}_2) - z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} \\ & < (p_1 - p_2) < \\ & (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} \end{aligned}$$